

NETRA-MAIL

From a suspicious email to an investigation-ready forensic report.

Problem Statement ID: 26106 — confirm

PS Title: [Exact portal title to be confirmed]

Theme: [Confirm portal theme]

PS Category: Software

Team ID: [To be added]

Team Name: [Registered name to be added]

DETECT

Explainable phishing risk

INVESTIGATE

Message, URL and relay evidence

PRESERVE

Case timeline and forensic report

Proposed Solution (Describe your Idea/Solution/Prototype)

NETRA-Mail connects a Gmail risk badge to a complete SOC investigation.

Detailed explanation

One connected workflow

- Analyze the selected email.
- Explain sender, text and link risks.
- Open the matching forensic report.

How it addresses the problem

Evidence behind the alert

- Makes warning signals readable.
- Reduces fragmented investigation.
- Preserves context for follow-up.

Innovation and uniqueness

Evidence-aware decisions

- Labels missing authentication data.
- Connects related indicators.
- Combines risk, trace and custody.

Analyze current email → Risk + reasons → View forensic report → Analyst action

Methodology and process for implementation

01 INPUT

Manual Gmail scan
or original EML

02 ANALYZE

Text + ML + URLs
headers +
attachments

03 EXPLAIN

Risk + findings
confidence + limits

04 TRACE

Relay / IP context
related indicators

05 PRESERVE

Case + SHA-256
report + custody

Evidence gate: original bytes enable signature checks; reconstructed Gmail content is labelled incomplete.

Technologies to be used

INTERFACE

JavaScript MV3
Streamlit dashboard

API + INTELLIGENCE

Python · FastAPI
scikit-learn · DNS

EVIDENCE

SQLite · SHA-256
AES-GCM encryption

DEPLOYMENT

Render API + dashboard
HTTPS · scoped API keys

Analysis of the feasibility of the idea

Technical feasibility

Working Gmail extension, FastAPI analysis,
Streamlit investigations and report generation.

Operational and financial viability

Pilot within one institution; open-source stack.
Hosting, storage and intelligence have running costs.

149 automated tests passed

Local regression evidence; not detection accuracy.

Potential challenges and risks → Strategies

Incomplete message headers

Fetch original MIME / upload EML; label unavailable checks.

False positives and model drift

Analyst review; evaluate on unseen data before rollout.

Privacy and online access

Scoped identities, encryption, retention and dashboard login.

Provider outages and persistence

Show unavailable signals; durable storage and backups.

Potential impact on the target audience

EMAIL USERS

Understand why a message is risky and open its supporting report.

SOC / IT TEAMS

Triage findings in one place; correlate recurring indicators.

INSTITUTIONS

Build consistent incident records for review and response.

Benefits of the solution (social, economic, environmental, etc.)

SOCIAL

Clear warnings support phishing awareness and informed decisions.

ECONOMIC

Aim to reduce review effort and fraud exposure; validate in pilots.

RESOURCE USE

Digital reports reduce paper use; manual scans limit unnecessary work.

Pilot success measures: false-positive rate • phishing recall • review time • report completeness

Details / Links of the reference and research work

01 Email authentication

Original-byte integrity and signing-domain checks.

[RFC 6376 — DKIM signatures](#)

02 Sender authorization

SPF needs trusted SMTP connection context.

[RFC 7208 — Sender Policy Framework](#)

03 Domain alignment

Connect authentication evidence to the From domain.

[RFC 7489 — DMARC](#)

04 Forensic methodology

Collection, examination, analysis and reporting.

[NIST SP 800-86](#)

Prototype and validation evidence

[NETRA-MAIL source repository • implementation, tests and deployment setup](#)